

STA 2002

Lecture 5

Confidence interval for a sample mean
Confidence interval for the difference between two means

Review

An estimator $U = u(X_1, X_2, \cdots, X_n)$ of θ is a statistics

Maximum likelihood estimator

$$\hat{\theta}_{MLE} = \arg\max L(\theta | X_1, X_2, \cdots, X_n)$$

Method of moment estimator: If $\theta = f(\mu_1, \mu_2, \cdots)$, then

$$\hat{\theta}_{MME} = f(\hat{\mu}_1, \hat{\mu}_2, \cdots)$$

Good estimators

- Unbiased
- Small variance
- Easy to compute

Definition

Confidence interval for a population parameter θ :

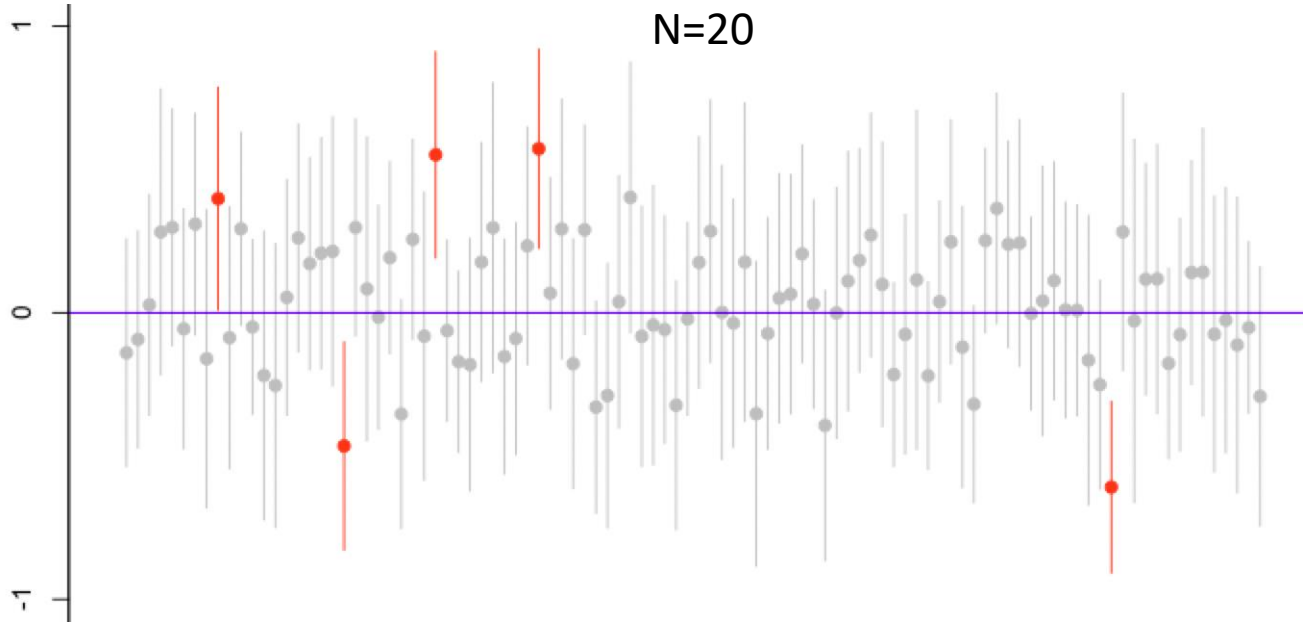
A random interval calculated from the sample, that **contains θ with some specified probability, called coverage probability**.

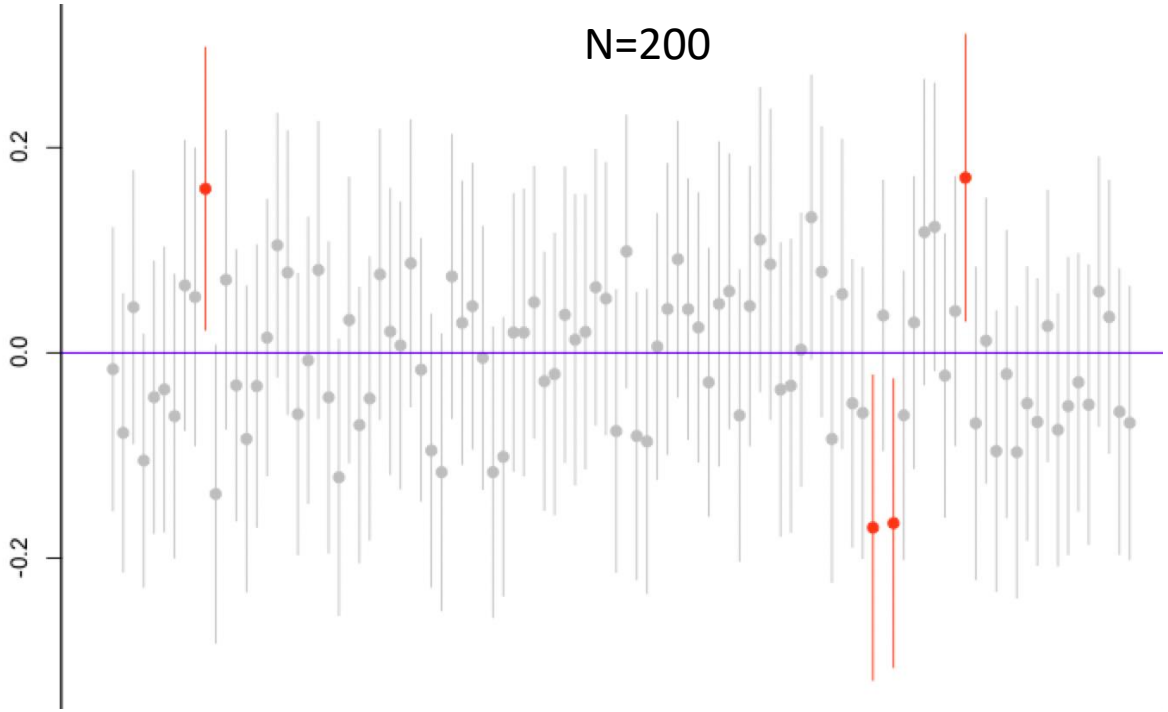
Confidence intervals are used in **conjunction with point estimators** to convey information about the uncertainty of the estimators.

If the coverage probability is $1-\alpha$, the interval is called a $100(1-\alpha)\%$ confidence interval.

Ex: Let α=0.05, a 95% confidence interval for μ is a random interval that contains μ with probability 0.95.

Example





"A 95% confidence interval for μ is a random interval that contains μ with probability 0.95"

means

"If we repeatedly generate 100 samples from the population, then about 5 confidence intervals will not contain μ"

Definition

The construction of CI is based on distributions. Most commonly normal due to CLT.

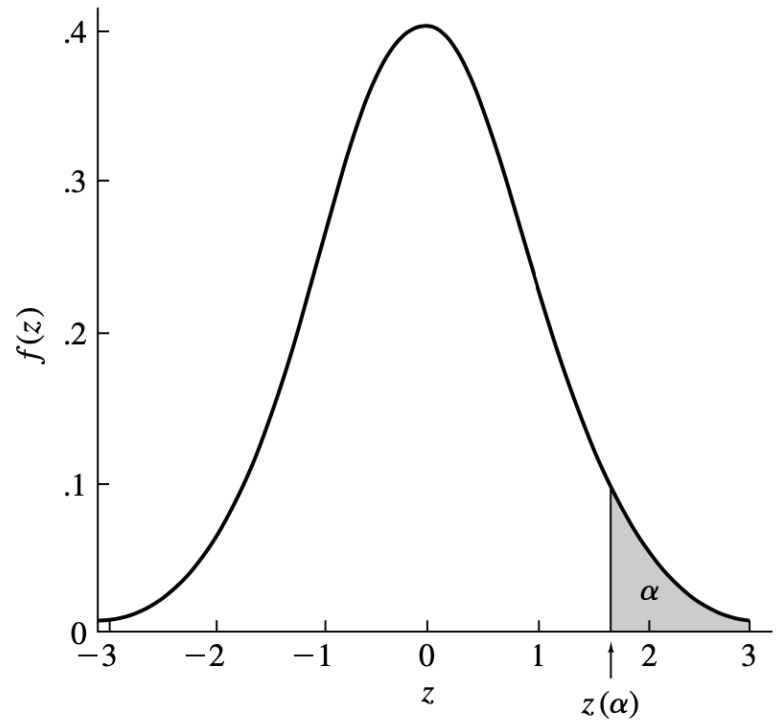


FIGURE **7.3** A standard normal density showing α and $z(\alpha)$.

Let $z(\alpha)$ be the number such that the area under the standard normal density function to the right of $z(\alpha)$ is α.

Then if $Z \sim N(0,1)$

$$P(-z(\alpha/2) \leq Z \leq z(\alpha/2)) = 1 - \alpha$$

Definition

$$P(-z(\alpha/2) \leq Z \leq z(\alpha/2)) = 1 - \alpha$$

Based on CLT, $(\overline{X} - \mu)/\sigma_{\overline{X}}$ approximately follows a standard normal distribution. So,

$$P\left(-z(\alpha/2) \leq \frac{\overline{X} - \mu}{\sigma_{\overline{X}}} \leq z(\alpha/2)\right) \approx 1 - \alpha$$

Rearrange,

$$P(\overline{X} - z(\alpha/2)\sigma_{\overline{X}} \leq \mu \leq \overline{X} + z(\alpha/2)\sigma_{\overline{X}}) \approx 1 - \alpha$$

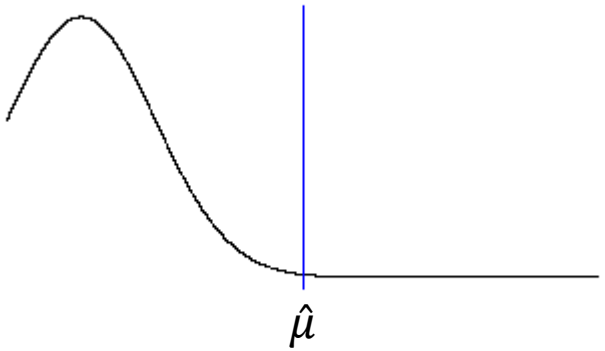
This is the basis for a two-sided CI

The above is true for every α, then we are treating μ as if it is from a distribution centered at $\overline{X}$

Another way thinking about this

From the data, we calculated the point estimator $\hat{\mu}$ of μ

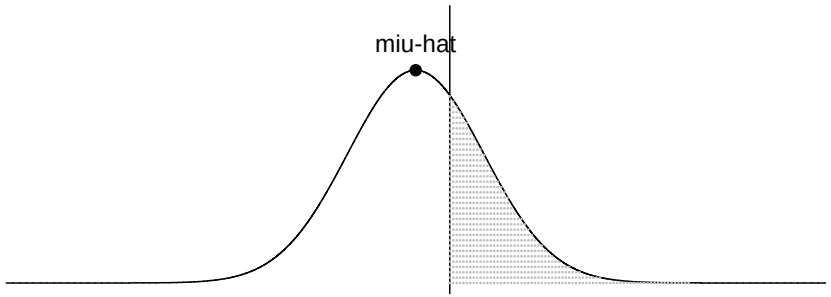
Q: what is the likely values of μ that allowed us to observe $\hat{\mu}$?

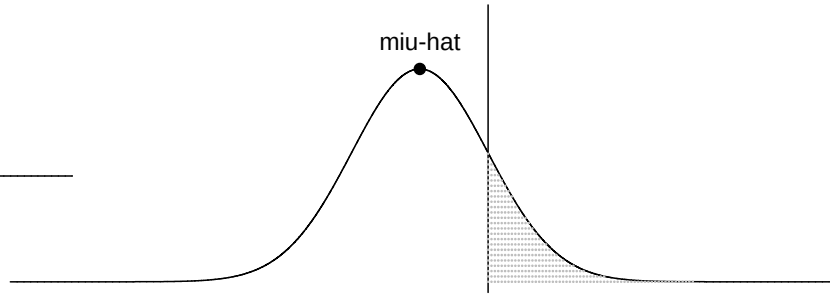


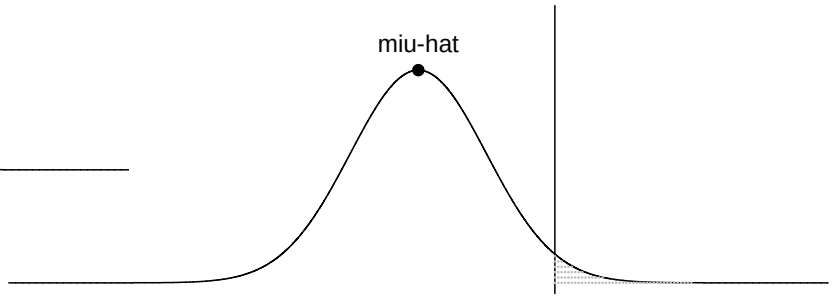
Using normal mean estimation as an example:

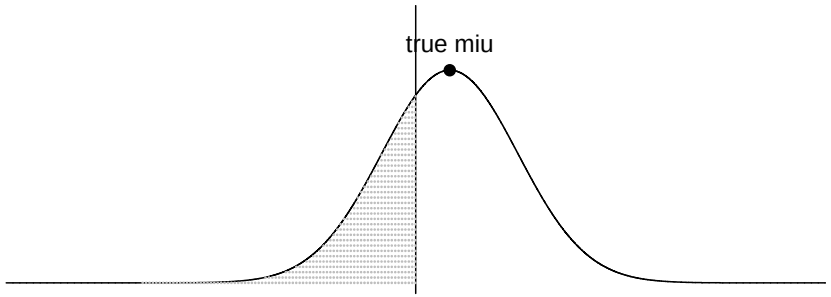
- Suppose we know the true σ. We want to find μ
- We have a sample size *n*, and sample mean $\bar{x}$
- Following MLE, we set $\hat{\mu} = \bar{x}$
- Based on theory about normal distribution, we know $\hat{\mu} \sim N(\mu, \frac{\sigma^2}{n})$
- What is the "distribution" of μ given we know $\hat{\mu}$?

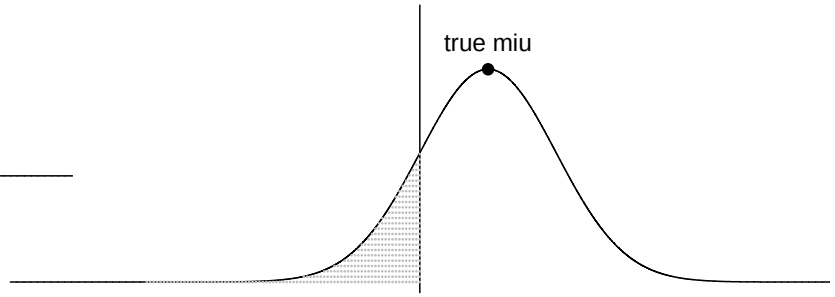
Another way thinking about this

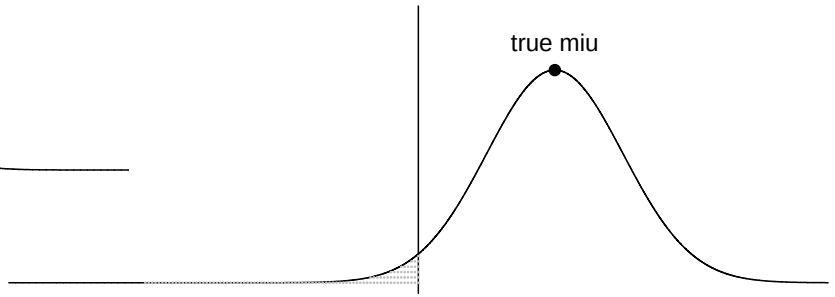












Another way thinking about this

Given we observe $\hat{\mu}$, the value of true μ is more likely to be close to $\hat{\mu}$. The further away from $\hat{\mu}$, the less likely.

The shape of the likelihood is also normal, based on the symmetric relation we saw in the last page:

$$N(\hat{\mu}, \frac{\sigma^2}{n})$$

Note:

μ is hidden truth.
So this is not a distribution of a random variable called μ.
It is just a function that reflects our belief about μ, given a sample.

CI for the mean (1): X is normally distributed, σ is known

$\hat{\mu}$:	$\bar{X}$
s.e. of $\hat{\mu}$:	$\frac{\sigma}{\sqrt{n}}$
Distribution of $\hat{\mu}$:	Exact normal, $N(\mu, \frac{\sigma^2}{n})$

The $100(1-\alpha)\%$ confidence interval:

$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

CI for the mean (2): X is arbitrarily distributed, σ is known

$\hat{\mu}$:	$\bar{X}$
s.e. of $\hat{\mu}$:	$\frac{\sigma}{\sqrt{n}}$
Distribution of $\hat{\mu}$:	Approximately normal, $N(\mu, \frac{\sigma^2}{n})$

The $100(1-\alpha)\%$ confidence interval:

$$\left[\bar{x} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \bar{x} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

CI for the mean (3): X is normally distributed, σ is NOT known

$\hat{\mu}$:	$\bar{X}$
s.e. of $\hat{\mu}$:	$\frac{\sigma}{\sqrt{n}}$ $(unknown, best\ guess\ is \frac{s}{\sqrt{n}})$
Distribution of $\hat{\mu}$:	t_{n-1} after transformation (uncertainty in s.e. reflected in bigger tail than normal)

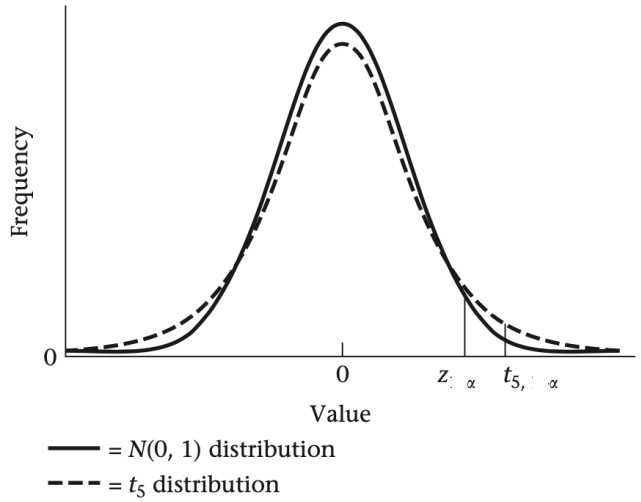
$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$T = \frac{\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

The $100(1-\alpha)\%$ confidence interval:

$$\left[\bar{x} - t_{\alpha/2(n-1)}\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2(n-1)}\left(\frac{s}{\sqrt{n}}\right)\right]$$

Figure 6.6 Comparison of Student's *t* distribution with 5 degrees of freedom with an *N*(0, 1) distribution



Compared to previous page: replace σ by *s*, replace the *z* statistic by a *t* statistic

CI for the mean (4): X is arbitrarily distributed, σ is NOT known

$\hat{\mu}$:	$\bar{X}$
s.e. of $\hat{\mu}$:	$\frac{\sigma}{\sqrt{n}}$ $\quad (unknown, best\ guess\ is \frac{s}{\sqrt{n}})$
Distribution of $\hat{\mu}$:	Approximately t_{n-1} after transformation (uncertainty in s.e. reflected in bigger tail than normal)

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$T = \frac{\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}\Big/(n-1)}} = \frac{\overline{X}-\mu}{S/\sqrt{n}} \sim t_{n-1} \quad \text{approximately}$$

The $100(1-\alpha)\%$ confidence interval:

$$\left[\bar{x} - t_{\alpha/2}(n\text{–}1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n\text{–}1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

CI for the mean: Summary

X is normally distributed, σ is known

$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

X is arbitrarily distributed, σ is known

$$\left[\bar{x} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \bar{x} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

X is normally distributed, σ is NOT known

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

X is arbitrarily distributed, σ is NOT known

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right]$$

Table 6.3 Five random samples of size 10 from the population of infants whose birthweights (oz) appear in Table 6.2

	Sample				
Individual	1	2	3	4	5
1	97	177	97	101	137
2	117	198	125	114	118
3	140	107	62	79	78
4	78	99	120	120	129
5	99	104	132	115	87
6	148	121	135	117	110
7	108	148	118	106	106
8	135	133	137	86	116
9	126	126	126	110	140
10	121	115	118	119	98
$\bar{x}$	116.90	132.80	117.00	106.70	111.90
s	21.70	32.62	22.44	14.13	20.46

Compute a 95% CI for the mean birthweight based on the first sample of size 10 in Table 6.3.

We have $n = 10$, $\bar{x} = 116.90$, $s = 21.70$. Because we want a 95% CI, $\alpha = .05$. Therefore, from Equation 6.6 the 95% CI is

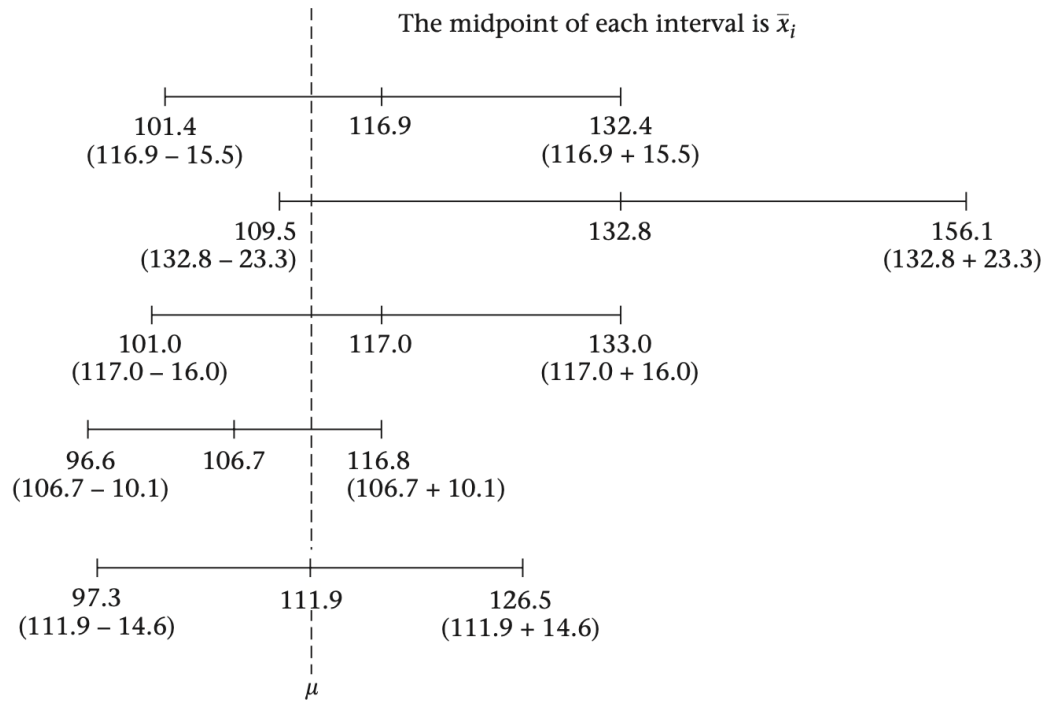
$t_{0.025}(9)=2.262$

$$\left[\bar{x} - t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right), \bar{x} + t_{\alpha/2}(n-1)\left(\frac{s}{\sqrt{n}}\right)\right] = \left[116.9 - 2.262(21.70)/\sqrt{10}, 116.9 + 2.262(21.70)/\sqrt{10}\right]$$
$$= (116.9 - 15.5, 116.9 + 15.5)$$
$$= (101.4, 132.4)$$

Table 6.3 Five random samples of size 10 from the population of infants whose birthweights (oz) appear in Table 6.2

	Sample				
Individual	1	2	3	4	5
1	97	177	97	101	137
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6	148	121	135	117	110
7	108	148	118	106	106
8	135	133	137	86	116
9	126	126	126	110	140
10	121	115	118	119	98
$\bar{x}$	116.90	132.80	117.00	106.70	111.90
s	21.70	32.62	22.44	14.13	20.46

Figure 6.7 A collection of 95% CIs for the mean μ as computed from repeated samples of size 10 (see Table 6.3) from the population of birthweights given in Table 6.2



Factors Affecting the Length of a CI

The length of a $100\% \times (1 - \alpha)$ CI for μ equals $2t_{n-1,\ \alpha/2}\, s/\sqrt{n}$ and is determined by n, s, and α.

- n — As the sample size (n) increases, the length of the CI decreases.
- s — As the standard deviation (s), which reflects the variability of the distribution of individual observations, increases, the length of the CI increases.
- α — As the confidence desired increases (α decreases), the length of the CI increases.

One-sided Confidence Interval (rare)

Ex: X is normally distributed, σ is known



$$\left[\overline{X} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right), \overline{X} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)\right]$$

$$[\bar{x} - z_{\alpha}(\sigma/\sqrt{n}), \infty)$$

$$(-\infty, \bar{x} + z_{\alpha}(\sigma/\sqrt{n})]$$

CI for the difference between two means (1): X, Y are normally distributed, σ_x and σ_y are known

$\hat{\mu}_x$	$\bar{X}$
s.e. of $\hat{\mu}_x$	$\frac{\sigma_x}{\sqrt{n}}$
Distribution of $\hat{\mu}_x$	Exact normal, $N(\mu_x, \frac{\sigma_x^2}{n})$

$\hat{\mu}_y$	$\bar{Y}$
s.e. of $\hat{\mu}_y$	$\frac{\sigma_y}{\sqrt{m}}$
Distribution of $\hat{\mu}_y$	Exact normal, $N(\mu_y, \frac{\sigma_y^2}{m})$

	$\bar{X} - \bar{Y}$
s.e.	$\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}$
Distribution of $\bar{X} - \bar{Y}$	Exact normal, $N(\mu_x - \mu_y, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$

The $100(1-\alpha)\%$ confidence interval:

$$[\bar{x} - \bar{y} - z_{\alpha/2}\sigma_W, \bar{x} - \bar{y} + z_{\alpha/2}\sigma_W]$$

$$\sigma_W = \sqrt{\sigma_X^2/n + \sigma_Y^2/m}$$

CI for the difference between two means (2.1): X, Y are normally distributed, σ_x and σ_y are NOT known

If n and m are large, s is close to σ. Do a simple replacement. The $100(1-\alpha)\%$ confidence interval:

$$\bar{x} - \bar{y} \pm z_{\alpha/2}\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

CI for the difference between two means (2.2): X, Y are normally distributed, σ_x and σ_y are NOT known

If m and n are not large (say, considerably smaller than 30), and if we can assume the two groups have the same variance

$$\sigma_X^2 = \sigma_Y^2 = \sigma^2$$

The pooled variance (to stabilize the variance estimate)

$$U = \frac{(n-1)S_X^2}{\sigma^2} + \frac{(m-1)S_Y^2}{\sigma^2} \quad \sim \quad \chi^2(n+m-2)$$

$$T = \frac{Z}{\sqrt{U/(n+m-2)}} \quad \sim \quad t_{n+m-2}$$

$$T = \frac{\dfrac{\overline{X} - \overline{Y} - (\mu_X - \mu_Y)}{\sqrt{\sigma^2/n + \sigma^2/m}}}{\sqrt{\left[\dfrac{(n-1)S_X^2}{\sigma^2} + \dfrac{(m-1)S_Y^2}{\sigma^2}\right] \Big/ (n+m-2)}} \quad \sim \quad t_{n+m-2}$$

CI for thedifference between two means (2.2): X, Y are normally distributed, σ_x and σ_y are NOT known

$$\frac{\overline{X} - \overline{Y} - (\mu_X - \mu_Y)}{\sqrt{\left[\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}\right]\left[\frac{1}{n} + \frac{1}{m}\right]}} \quad \sim \quad t_{n+m-2}$$

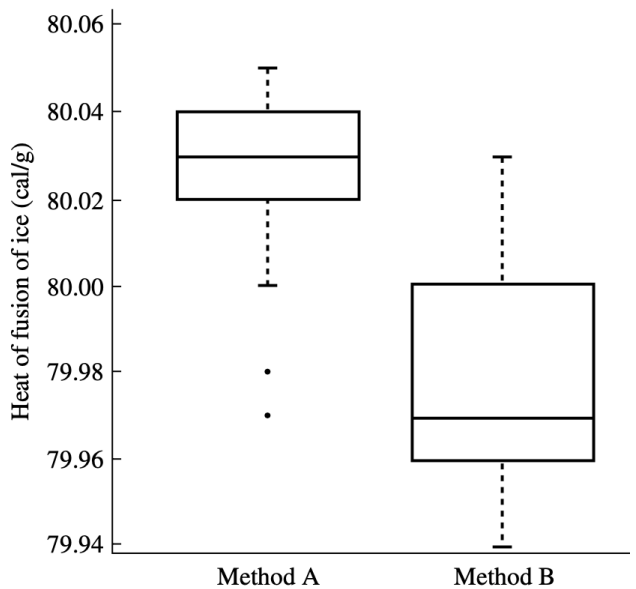
The CI is

$$\left[\bar{x} - \bar{y} - t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}, \bar{x} - \bar{y} + t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}\right]$$

$$t_0 = t_{\alpha/2}(n+m-2)$$

$$S_P = \sqrt{\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}}$$

Method A	Method B
79.98	80.02
80.04	79.94
80.02	79.98
80.04	79.97
80.03	79.97
80.03	80.03
80.04	79.95
79.97	79.97
80.05	
80.03	
80.02	
80.00	
80.02	



$$\overline{X}_A = 80.02 \qquad S_a = .024$$
$$\overline{X}_B = 79.98 \qquad S_b = .031$$

$$s_p^2 = \frac{12 \times S_a^2 + 7 \times S_b^2}{19} = .0007178$$
$$s_p = .027$$

$$s_{\overline{X}_A - \overline{X}_B} = s_p \sqrt{\frac{1}{13} + \frac{1}{8}}$$
$$= .012$$

$$\overline{X}_A - \overline{X}_B = .04$$

$$(\overline{X}_A - \overline{X}_B) \pm t_{19}(.025) s_{\overline{X}_A - \overline{X}_B} = (.015, .065)$$

CI for the difference between two means: Summary

Assume that X, Y are normally distributed,

- σ_x and σ_y are known

$$[\bar{x} - \bar{y} - z_{\alpha/2}\sigma_W, \bar{x} - \bar{y} + z_{\alpha/2}\sigma_W] \qquad \sigma_W = \sqrt{\sigma_X^2/n + \sigma_Y^2/m}$$

- σ_x and σ_y are NOT known and n and m are large

$$\bar{x} - \bar{y} \pm z_{\alpha/2}\sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

- σ_x and σ_y are NOT known, n and m are not large and $\sigma_x = \sigma_y$

$$\left[\bar{x} - \bar{y} - t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}, \bar{x} - \bar{y} + t_0 s_p\sqrt{\frac{1}{n} + \frac{1}{m}}\right]$$

$$t_0 = t_{\alpha/2}(n+m-2)$$

$$S_P = \sqrt{\frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}}$$